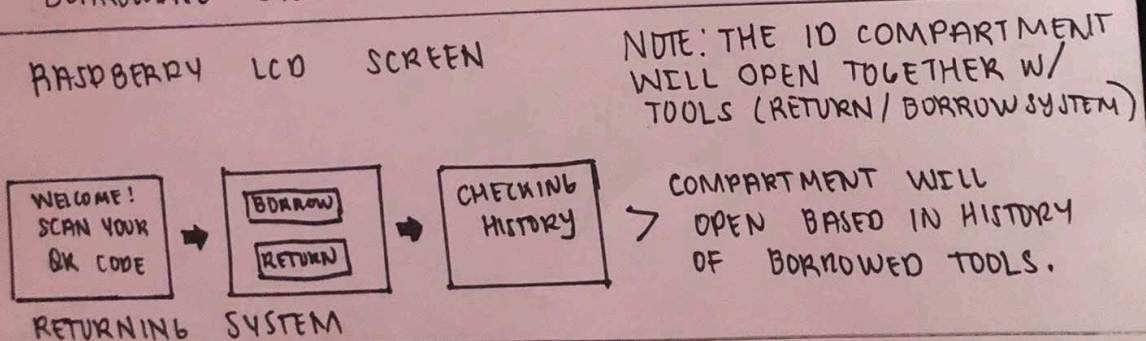
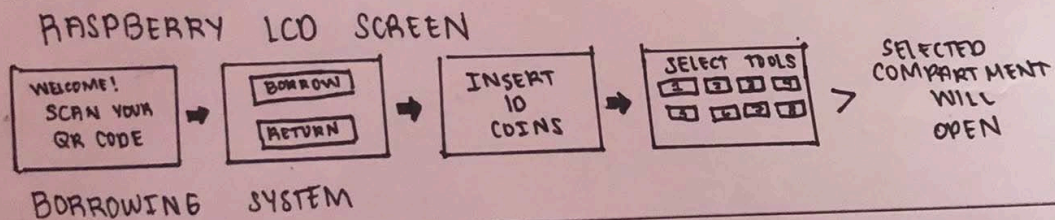
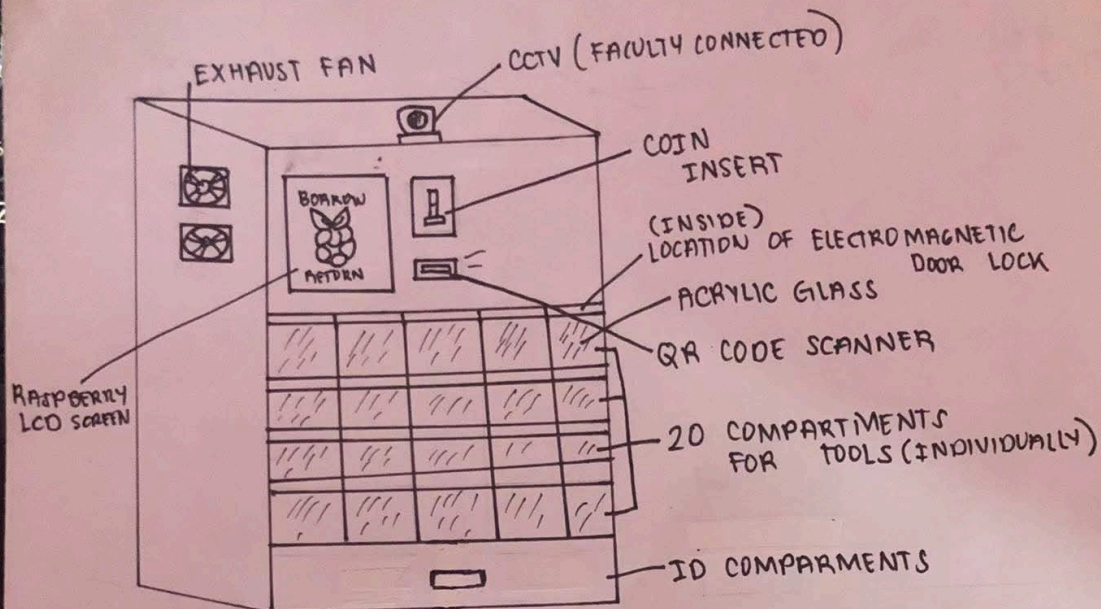


PROCESS WE WANT TO DEVELOP IN THE WEB AND ALSO THE MATERIALS WILL BE USED IN THE SYSTEM. HERE'S THE DESIGN OF THE PROTOTYPE FOR THE OVERVIEW:



TOOLS IN EACH COMPARTMENTS (20 PCS)

5 → CALCULATOR	5 → DUAL SCREWDRIVERS
5 → WIRE STRIPPER	5 → COMBINATION PLIERS

16 compartments _ 4:calculator, 4: multimeter, 4:dual screwdrivers, 4:combination pliers.

MATERIALS:

BRAIN OF THE SYSTEM

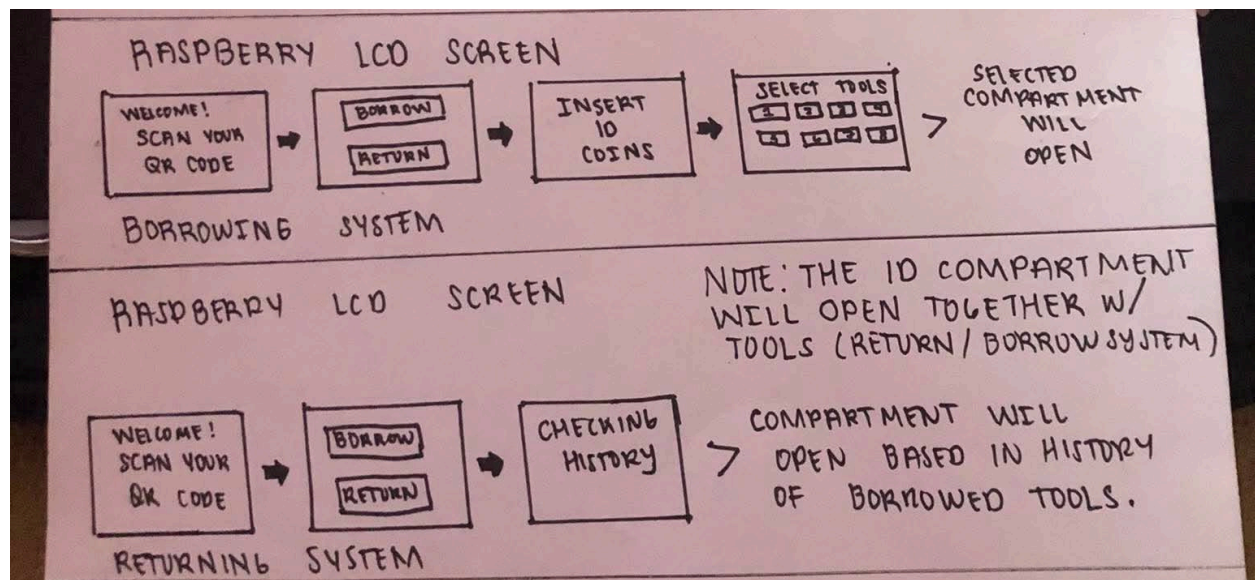
1. RASPBERRY PI 4 MODEL B
(DISPLAY WELCOME SCREEN FOR TRANSACTION)
2. ESP32 DEVKIT V1
(CONTROL ELECTROMAGNETIC(compartments), COIN ACCEPTOR AND SENDS ALERT TO THE ADMIN WHENEVER THE TOOLS NOT RETURN BY 3 HRS)

HARDWARE THAT CONNECTED IN THE SYSTEM WEB (PROBABLY IN ESP32 SYSTEM)

1. ELECTROMAGNETIC DOOR LOCK (20 COMPARTMENTS)
2. QR CODE SCANNER
3. COIN ACCEPTOR

PURPOSE: TO RECORD THE LOG IN AND LOG OUT OF THE STUDENTS AND TO MAINTAIN SAFETY OF THE RENTAL TOOLS.

WE EXPECT FOR THIS FLOW OF THE WEB:



BORROW SYSTEM:

1. Welcome Screen for QR Code Scanning
 - o Student scans QR Code.
 - o System verifies and records the Student TUPM Number in the Raspberry Pi database.
2. Select Command
 - o BORROW
 - o RETURN
3. Borrowing Eligibility Check
 - o System checks if the student currently has an unreturned tool.
 - o If none, proceed to the next step.

4. Insert Coin Deposit (₱10.00)

- Coin acceptor verifies the inserted coin.

5. Tool Selection Screen (4 types of tools)

- Display available tools.
- Display available 5 compartments each tools.
- Student selects the desired tool.

6. Tool Release Process

- Raspberry Pi sends a command to the ESP32.
- ESP32 deactivates the corresponding electromagnetic door lock.
- Selected tool compartment opens.
- Student ID compartment opens.

7. Transaction Recording

- System records:
 - Student TUPM Number
 - Borrowed Tool
 - Compartment Number
 - Date and Time Borrowed

8. Compartment Closure

- Student manually closes the tool compartment and ID compartment.
- Electromagnetic door locks are reactivated.

RETURN SYSTEM

1. Welcome Screen for QR Code Scanning

- Student scans QR Code.
- System verifies and records the Student TUPM Number.

2. Select Command

- BORROW
- RETURN

3. Borrowing Record Verification

- System retrieves the student's active borrowing record.

4. Compartment Identification

- System identifies:
 - Borrowed Tool Compartment
 - Student ID Compartment

5. Return Process

- Raspberry Pi sends a command to the ESP32.
- ESP32 deactivates the corresponding electromagnetic door locks.
- Tool compartment and ID compartment open.

6. Tool Return Confirmation

- Student place the borrowed tool inside the designated compartment.
- Student retrieves the ID card.

7. Transaction Update

- System records:
 - Date and Time Returned
 - Tool Status (Available)
 - Transaction Completion

8. Compartment Closure

- Student manually closes the tool compartment and ID compartment.
- Electromagnetic door locks are reactivated.

MONITORING SYSTEM

1. Borrowing Transaction Monitoring

- Raspberry Pi continuously monitors active borrowing transactions stored in the database.

2. Time Limit Verification

- System compares the current time with the recorded borrowing time.
- Maximum borrowing duration is 3 hours.

3. Overdue Detection

- If a borrowed tool exceeds the 3-hour limit, the transaction is marked as overdue.

4. Administrator Notification

- System sends an alert to the administrator.
- Alert includes:
 - Student TUPM Number
 - Borrowed Tool
 - Borrowing Time
 - Overdue Duration

5. Dashboard Update

- Overdue transactions are displayed in the administrator monitoring dashboard.

NOTE: IF YOU HAVE ANY RECOMMENDATION WITH THIS FLOW JUST MESSAGE US TO THE GC TO LOOK FORWARD IN THE CHANGES ESPECIALLY IN THE NUMBER OF COMPARTMENTS, THANK YOU :3

Development of Automated Laboratory Tools Inventory Systems

A Project Study

Presented to the Faculty of
Electrical Engineering Technology Department
College of Industrial Technology
Technological University of the
Philippines-Manila
Ayala Boulevard, Manila

by

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In Partial Fulfillment of the Requirements for
the Degree **Bachelor of Engineering**
Technology major in Electrical Engineering
Technology

May 2026

ABSTRACT

In the world of industrial inventory systems, mistakes happen and the data recorded can be lost in an instant. This not only happens in large industries but as well as small scale inventory systems. The researchers formulated a solution to the existing problem and to innovate the already existing inventory systems. To elaborate the Research GAP that the researchers found, some systems in inventory solely rely on manual writing of data, and sometimes the data written in the paper lost due to negligence of the person assigned to take the data and will also lead to loss of assets and increase in liability when it comes to inventory items. In the Technological University of The Philippines - Manila, researchers conducted a system that does not rely on manual data collection, hence, Developed an Automated Laboratory Tools Inventory System, which will focus only on laboratory tools that are being used in laboratory activities. This system helps to record and to know who will be the one using the tools with a certain amount of time given from the moment they remove the tools from the inventory, by this system the beneficiaries and the researchers will not worry about losing their data because it is recorded whatever happens in the inventory system. For added features of this project, researchers will use the microcontroller Raspberry Pi 4 model B and ESP32 for the main component which will act as "brain" and "limbs" of the system.

Chapter 1

INTRODUCTION

Background of the Study

Inventory systems are essentials in industrial application, it helps to sort heavy machinery and light tools. Sorting of tools and equipment might increase efficiency and productivity of a company. Without a proper inventory system, companies may face challenges in tracking their equipment, where it has been, who used it last, where did it go, what the status of the equipment, is it broken or not, and many other several questions that might decrease the productivity of the company. Proper management systems for tools will increase the organization to mitigate such things as sorting machines and equipment. Some management systems include barcode scanning, QR code scanning, and other automations. By automating these processes, companies can reduce the time and resources needed to manage their inventory manually. In the tool inventory system, it tracks the status of tools depending on its environment, as well as its location. In this kind of inventory system, it measures the accurate record of the tools, it reduces downtime and improves productivity and saves business money and repair costs. (C. Weber., 2023).

In a similar context, managing the laboratory tools within engineering classrooms is another essential task, especially within classrooms that include electrical engineering students who use specific laboratory tools to complete their activities. Similar to the way in which industrial environments manage the tools necessary to complete their tasks, educational laboratories must also manage their tools.

However, many educational facilities still utilize manual systems to manage their laboratory tools inventory.

These manual systems have the potential to lead to errors in managing these tools and can create challenges for the educators to effectively manage their classrooms. Studies have demonstrated that manual inventory systems for tools lack the accuracy and efficiency of automated systems for tool management within classrooms, leading to the poor management of these resources in general.(Aguila & Balba, 2020).

The idea about the inventory of tools is to help the electrical technology students to support in their needs while having an electrical laboratory in the lobby, The problem of electrical engineering students that they do not have proper place for their tools, Implementing their needs and to be encouraged to use this in order if the student cannot afford tools, students can borrow on it and after their time limit they should return it in inventory it will automatically remind the one who is in charge in faculty room whom did not return. Our Goal is to also help the Professors. They can also use that if they need to borrow it. This increases the likelihood of tool loss, misuse, and inefficient management of laboratory resources (Rahman et al., 2018).

Various studies reveal that the use of barcode systems, RFID technologies, and other web based inventory systems improves the management of tools when it comes to tools inventory systems. In some systems that stated above, it also comes with advantages and disadvantages. Some advantages of it are the collaboration of hardware and software systems,

They share the same goal and command to function properly depending on the assigned command. In some ways it has certain limitations when it comes to finances, because some materials are expensive due to its electronic nature.

Other systems only focus on user identification without confirming if the tool used is in a good condition or not, others do not incorporate their system with storage protection programs. This limitation shows the gap in developing comprehensive systems that use some user authentication systems, tracking of tools in real time and other features such as security systems. (Ayomide et al., 2024).

To address these concerns, The researchers propose the development of an automated laboratory tool inventory system that incorporates QR code technology for real time record of tool borrowing and return. It aims to prevent loss and facilitate the identification of students who have borrowed and not returned within the allotted time limit that incorporates each tool. The QR code will be scanned by the programmable system such as Raspberry Pi 4 Model B and ESP32 Dev Kit V1 that will serve as the brain of the inventory. In addition, electromagnetic door locks are built inside the compartments to ensure the safety of the tools, ensuring accurate tracking and security. This approach improves the reliability of inventory management and supports the implementation of automated and real time tracking systems of the tools, which have been proven to significantly improve tracking accuracy and operational performance (Erlangga et al., 2022).

Objectives of the Study

The development of the automated laboratory tools inventory system aims to provide an efficient and secure solution for managing laboratory tools and equipment.

Specifically, it aimed to:

1. Designed as follows:
 - a) It utilizes QR code technology for user identification and tracking of borrowing and returning of tools.
 - b) It is integrated with embedded systems such as Raspberry Pi 4 Model B and ESP32 Dev Kit V1 for system control and communication.
 - c) It includes an electronic locking mechanism to ensure secure storage of laboratory tools.
 - d) It has a system interface that records and monitors user activity and inventory data in real time.
 - e.) it has a student ID compartment that temporarily stores the borrower's identification during tool rental transactions to enhance accountability and ensure proper tool return.
 - f.) it has a coin acceptor mechanism that requires users to deposit a specified amount before borrowing laboratory tools, promoting accountability and encouraging the timely return of borrowed equipment.
 - g.) it has a notification and monitoring system capable of tracking borrowed tools and alerting administrators regarding overdue tool returns.
2. Construct the prototype as designed.
3. Test and improve the prototype.

Scope and Limitations of the Study

This study mainly focuses on the design, development and implementation of an automated laboratory tools inventory system built with a smart cabinet for secure storage and monitoring of the tools. The researchers utilized QR code technology and programming systems, together with the Raspberry Pi 4 Model B and ESP32 Dev Kit V1, along with electro magnetic door lock. The system has a QR code scanner capable of scanning student QR codes for tools intended to record their borrowing and returning activities, and detecting tool availability within the cabinet compartments. The researchers developed two components, the hardware and software to ensure system functionality and effectiveness.

For the testing, the researchers evaluated the performance of the system in terms of response time, accuracy and reliability in tracking tool usage and user activity. The system was tested by actual laboratory scenarios involving students borrowing tools and laboratory personnel as users. The collected data were analyzed to determine the effectiveness of the system compared to traditional inventory methods that are manually listed with ink in the ordinary record book. Feedback and recommendations from electrical technology students were also gathered to further future improvements of the system's usability and effectiveness.

However, there are some technical limitations in the system including the QR code scanning feature that depends on the designated QR code in the compartment of the tools and it may not be compatible with all existing institutional identification systems. The system may only work if there is a wifi connection, it cannot be used when the internet is

down, especially in the ESP32 Dev Kit V1 that needs internet connections to receive and send notifications for warning purposes. The system focuses only on inventory management. It is designed specifically for the tools that will be used by the electrical engineering technology students. It is not a general inventory system that can be used for warehouses or retail stores and has a limit of 20 compartments, each compartments have 4 different types of tools intended to be easily detected by raspberry. The system can generate reports but it cannot predict future needs. The system is tested only within the university only.

This study promotes smart systems development that is aligned with the innovation of the Technological University of the Philippines. It also intends to support the United Nations Sustainable Development Goals, particularly Goal 9 (Industry, Innovation, and Infrastructure), by implementing the development of efficient, sustainable, and technology driven solutions in educational environments.

Chapter 2

REVIEW OF RELATED LITERATURE

This section discusses the details of the main topic, the problem and its root causes, and other pertinent information that provides deep theoretical or conceptual and practical underpinnings of the capstone study

In the Philippines academic laboratories now transition from traditional manual inventory methods to digital inventory systems which use automated and IoT technology. According to, (Bautista et al., 2023) shows that local engineering laboratories continue to use traditional logbook inventory systems instead of modern inventory systems. The laboratory operations and accountability systems face negative impacts from human mistakes and slow data updates and system errors, which create problems for laboratory operations.

To address these limitations, modern systems develop solutions to their existing problems through structured processes which resemble smart inventory systems that Philippine universities currently test. The system design includes four main operational stages which begin with data collection and data processing and data authentication and end with continuous system monitoring to achieve both higher operational accuracy and better system performance.

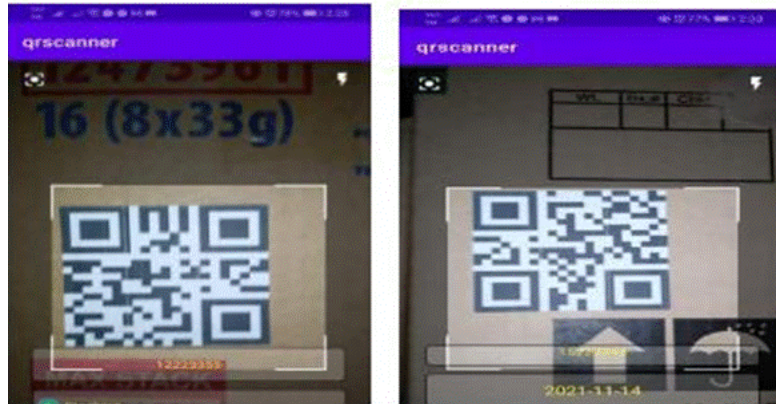


Figure 1: QR Code Based Inventory Authentication System

Source: https://www.ijaemr.com/uploads/pdf/archivepdf/2023/IJAEMR_578.pdf

This figure 1 shows that the QR code is embedded with identification data for each laboratory user. The Philippine universities use this attendance tracking method together with access control systems. (Reyes and Dela Cruz., 2022) describe how QR code technology improves operational efficiency through its user identification system which operates without physical contact.



Figure 2: Raspberry Pi Based Smart Inventory Controller

Source: <https://www.instructables.com/Integrated-Inventory-Management-System/>

In this figure 2, Users scan their QR codes using a camera module connected to the system. The system requires scanned data verification before users can access laboratory equipment. The authentication method used here complies with security

standards for smart campus projects in the Philippines which use IoT technology to restrict equipment access to authorized users only.

Existing Systems

QR code systems are widely used in inventory tracking and access control applications. The systems employ camera scanners for decoding information which they use to create a data record that enables monitoring activities. Real-time user activity tracking together with authentication verification uses QR scanning according to the findings from an IoT based QR system study conducted by researchers.



Figure 3: QR Code Based Inventory Authentication System

Source: https://www.ijaemr.com/uploads/pdf/archivepdf/2023/IJAEMR_578.pdf

As shown in figure 3, QR code system used for authentication and tracking. The system enables quick user identification while it documents all transactions because it needs operators to perform manual scanning which creates delays during extensive operations.

Microcontroller based IoT inventory systems have gained significant traction in contemporary applications. The IoT inventory system developed by (S. Jayanth et al.,

2017) employs Raspberry Pi devices and sensors to track inventory levels which automatically send stock updates through online platforms to users. The system cuts down on workforce requirements while it delivers better operational results.



Figure 4 ESP32 Barcode Inventory System

Source: <https://ojs.unikom.ac.id/index.php/injuratech/article/view/19200>

Figure 4 shows a barcode inventory system using ESP32 microcontroller for real time data transmission. The system enhances operational performance through rapid item detection but it requires operators to perform manual item scanning tasks.

RFID based inventory systems operate through their ability to detect items without using scanning processes. The research by (Bousselmi et al., 2024) demonstrates how RFID technology enables businesses to identify inventory items through radio waves which work without needing physical contact.

Smart Mirror using Raspberry Pi

According to Bhuvaneswari T, Aishwarya C & Aishwarya H A, B Abhinaya Nalina H D (2020) Raspberry Pi is not only applicable in various devices which includes Robotics and such other related devices, it enhances the use of raspberry Pi to its

futuristic idea. Integration of Raspberry Pi within a mirror helps people to do multi task and time checking at the same time.



Figure 5: Smart Mirror using Raspberry Pi

Source: <https://www.raspberrypi.com/tutorials/how-to-build-a-super-slim-smart-mirror>

Figure 5. The example of a smart mirror, they did not just put it in the mirror, they also put it in the everyday life of a person which helps them to look at themselves and be aware of the time, music, and even news at the same time.

Review of Related Literature and Studies

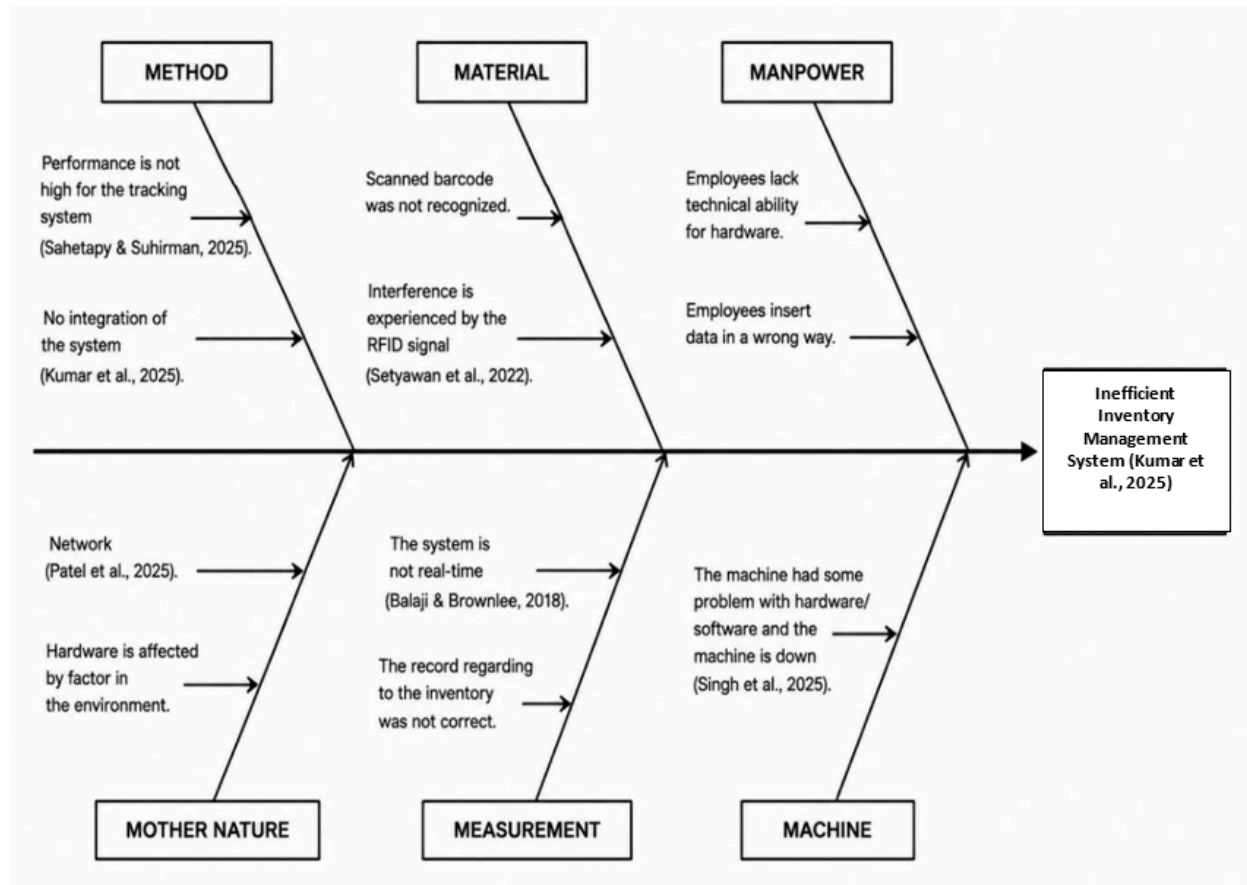
The literature review demonstrates that automated system operations experience multiple operational difficulties which emerge from the technological limitations of automated systems. (Setyawan et al., 2022) found that an inventory system which heavily relies on RFID and IoT technologies would face practical challenges because of signal interference and network problems and infrastructure requirements. The inventory management system control effectiveness will be adversely impacted by a system which operates through ineffective inventory processes and random operations and collects data incorrectly and manages data unsafely (Balaji & Brownlee, 2018). Monitoring and tracing equipment becomes problematic when the inventory manager lacks training and knowledge about old laboratory equipment and technology (Perez, 2023). The research

demonstrates that organizations require an advanced automated laboratory equipment inventory management system which provides both improved performance and enhanced scalability and simplification.

The automated inventory system in laboratories currently delivers real time tracking and monitoring accuracy and automated operation as its main benefits but laboratories and research institutions face two major challenges which include high establishment costs (Kumar et al., 2025) and security threats from system hacking and difficulties in system upkeep and system compatibility with other systems (Patel et al., 2025). The research presents an automatic laboratory equipment inventory management system which delivers cost-effective security and scalability as its main advantages while enhancing laboratory inventory control through advanced tracking and monitoring technology.

Figure 6

The Cause And Effect Model



The researchers conducted an extensive literature review to determine the strengths, weaknesses, opportunities, and threats (SWOT) of existing technological solutions, which served as a reference in the design and development of the current capstone study. Shown in Table 1 is the summary of the SWOT analysis, taken from published refereed sources.

Table 1

Research Title	Methodology	Strengths	Weaknesses	Opportunities	Threats
Role-Based Barcode System for Digitized Inventory Management in Engineering Applications (Sahetapy, & Suhirman, 2025)	Barcode based <u>role controlled</u> inventory system	Provide highest accuracy and highest accountability of tracking inventory with role-based security access	Provide highest accuracy and highest accountability of tracking inventory with role-based security access	May connect with mobile barcode scanning and IOT based inventory management	Unauthorized barcode scanning leading to unauthorized or duplication of barcode compromising the data integrity
Real-Time Asset Tracking and Management Using IoT, and RFID (Kumar et al., 2025)	Hybrid IoT, RFID, and AI-based tracking framework	Real-time monitoring and intelligence asset management	System complexity and requires integration of various interconnected technologies	Integration into predictive analysis, smart lab systems, etc.	Cyber-threat on inter-connected IoT systems
Development of Automatic Real-Time Inventory Monitoring System Using RFID Technology in Warehouse (Erlangga Bayu Setyawan, Ajeng Yunita, & Sekariatiningrum Rasmaydiwa Satriana, 2022)	RFID-based real-time warehouse inventory monitoring system	Real-time inventory tracking, efficiency, and stock monitoring accuracy	Requires stable RFID infrastructure, can be high on implementation costs	Expansion to IoT-based smart warehouse, automated inventory optimization	Cyber security issue, RFID interference and signal loss

Research Title	Methodology	Strengths	Weaknesses	Opportunities	Threats
Review of Laboratory Information Management System Design and Deployment (Singh et al., 2025)	Systematic review of laboratory information management system design and deployment in laboratory	Data management, compliance, and lab efficiency in its workflow work flow efficiency	Data management, compliance, and lab efficiency in its workflow	Requires significant infrastructure, can be high on maintenance cost	Development of cloud-based, hybrid LIMS Cybersecurity threat and system downtime
Real Time Inventory Tracking and Optimization Using IoT and Cloud Computing (Patel et al., 2025)	IoT and <u>cloud</u> based inventory optimization system	Centralized monitoring and real-time inventory update	Depends heavily on the Internet and cloud infrastructure	Development of automated smart inventory ecosystems	Network failure and cyber attacks

Figure 6

This Role Based Barcode System for digitized inventory management for engineering applications (Sahetapy & Suhirman, 2025) system as shown in Figure 6 function. This system leverages barcode technology and role based access control system in an attempt to boost the efficiency and integrity of inventory system. Role assignment to scanners or handlers are provided by the system, via clearly organized inventory system, which permits an effective control on inventory management. Issues where scanner made errors are caused by printer to produces low quality print out, and errors in the barcode labels, to which personnel must scan. The forthcoming system will feature the use of mobile scanning application to interlink with IoT based inventory systems, where the use of scanners is improving over time. Unauthorized scanning of barcode, along with barcode replication are the risk that can compromise the system security.



Figure 6

*Role Based Barcode System for Digitized Inventory Management in Engineering Applications
(Sahetapy & Suhirman, 2025)*

Figure 7

in Figure 7 Kumar et al. (2025) show their study; which illustrated their system called "Real-Time Asset Tracking and Management System Using IoT, RFID and Artificial Intelligence". In their system, both

IoT devices and RFID technologies are integrated with artificial intelligence, to realize both real-time tracking and decision support for asset management. The system provide automatic asset tracking functions that improves working efficiency for asset tracking. By the interaction among many different technologies, more difficulties on system integration and design are introduced, complex connected systems leads to security risks.



Figure 8

Development of Automatic Real Time Inventory Monitoring System Using RFID Technology in Warehouse (Setyawan et al., 2022)

Figure 9

Figure 9 illustrates the results from the Review of Laboratory Information Management System LIMS Design and Deployment for Laboratory Applications (Singh et al., 2025). The study emphasizes software based laboratory management systems, and helps organizations manage the processing of data and stay in accordance with compliance regulations, and improve operational efficiency. LIMS implementation in laboratories organizes workflow and reduce error prone tasks that can be due to manual data handling processes. However the LIMS requires vast resources and constant maintenance. Future markets will expand opportunity to cloud based LIMS and hybrid LIMS.



Figure 9

Laboratory Information Management System Design and Deployment (Singh et al., 2025)

Figure 10

Figure 10 shows the Real Time Inventory Tracking and Optimization System Using IoT and Cloud Computing system proposed by Patel et al. (2025). This system utilizes IoT devices and cloud computing to provide center access for control of inventory, and to always maintain know of your run rate at any given point in time. The system would bring about automation in inventory management, and better distribution of data, and efficiency in operations. The system requires functioning cloud computing services, and internet access to run. Future works in this system could involve fully automated smart inventory systems.

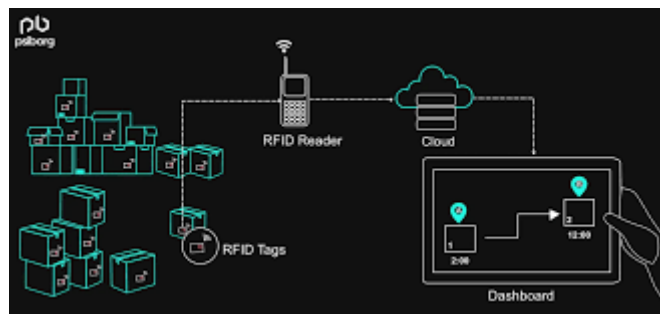


Figure 10

Real-Time Inventory Tracking and Optimization Using IoT and Cloud Computing (Patel et al., 2025)

Table 2
Related Patents

Patent Title	Author/Source	Adopted Feature	Enhanced Feature
System and Method for Automated Laboratory Inventory Management Using RFID Technology	(Jones et al., 2020)	A computer readable memory medium comprising program instructions for graphically developing a connectivity driver	Information stored in an RFID tag of a tagged item received at the delivery location
Object Tracking and Management System and Method Using Radio-Frequency Identification Tags	(Francis et al., 2003)	Location-based tracking identification system using RFIDs	Enables recording of the specific location of tagged items
Method and System for Inventory Control	(Clandatus et al., 2007)	Ability to determine whether an item in the inventory has been returned or not	Alert system that notifies when assets in the inventory are missing or not returned within a specified time
Monitoring Removal and Replacement of Tools Within an Inventory Control System	(Phillips et al., 2023)	Formation of an image of inventory tools using a computer system	The system can recognize the presence or absence of tools in real time
System and Method for Inventory Sharing in a Laboratory Management System	(Barnes et al., 2023)	A system for sharing consumable inventory in a laboratory management system	Includes middleware software controlled by a processor, multiple instruments connected to the middleware, and consumable items that can be removably installed in selected instruments

Figure 11

Figure 11 an RFID automated laboratory inventory management system (Jones et al. 2020). This laboratory inventory management system has embedded in computer readable memory medium program instructions used for the graphical development of connectivity driver for inventory

management and tracing. Each laboratory item has RFID tag for storing details like identity, status, and location of delivery. When the goods arrive in delivery place the RFID tag is scanned automatically by the machine.

Figure 11

System and Method for Automated Laboratory Inventory Management Using RFID Technology (Jones et al., 2020)

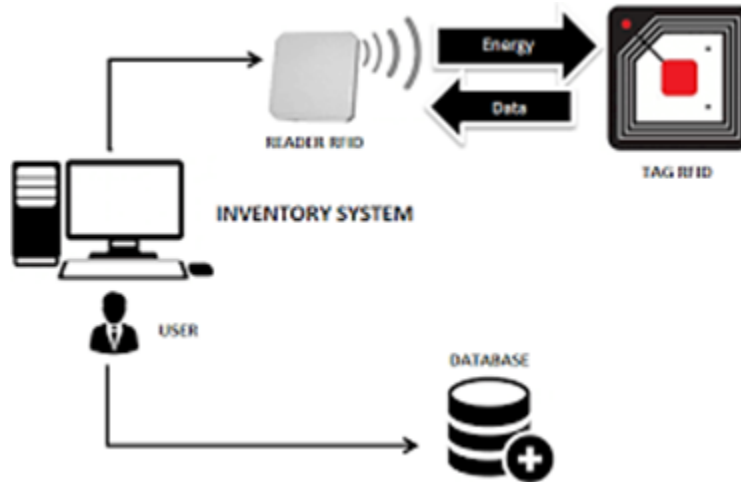


Figure 12

The diagram shown in Figure 12 is RFID based system for object tracking and management by Francis et al. (2003). This system basically provides an identification and tracking of objects based on the location using radio frequency identification tag. In this system, each and every object is associated with RFID tag to identify the location of the particular item in a particular environment. The RFID reader keeps scanning the objects with RFID tags in a specified environment and continuously updates the location information into the system database.

It is used to trace the location of the object in the specified environment. The added features to this system are the ability to trace the object and record spatial location, to overcome from lost and misplaced objects.

Figure 12

Object Tracking and Management System Using RFID Tags (Francis et al., 2003)



Figure 13

Figure 13 describes an inventory control system proposed by Claudatos et al. (2007). The system focuses on items presence in the system and on their return status in a inventory system. The system decides if a item has already returned to the system or is still in use. It automatically generate and sends an alert to user/administrator when an asset is absent and not returned in a predefined period. It enhances traditional inventory control by automatically alerting administrator when items is missed or overdue to return and consequently, reduces inventory loss and increase the responsibility of employees

Figure 13

Method and System for Inventory Control (Claudatos et al., 2007)



Figure 14

Figure 14 shows the developed tool replacement and removal monitoring system by Phillips et al. (2023). It applies the computer imaging technique to create a visual map for inventory tools. By using image recognition and monitoring algorithm, it is able to check the presence of tools in specific regions or whether it is vacant in that particular space to monitor the using/taking off/put back processes of the inventory tools. The improvement in this system is that it is capable of automatically identifying the inventory situation instead of scanning a code on the item and RFID; thus, it is applied for some situations that the visual inspection is more appropriate for the safety of the tools in inventory.

Figure14

Monitoring Removal and Replacement of Tools within Inventory Control System (Phillips et al., 2023)



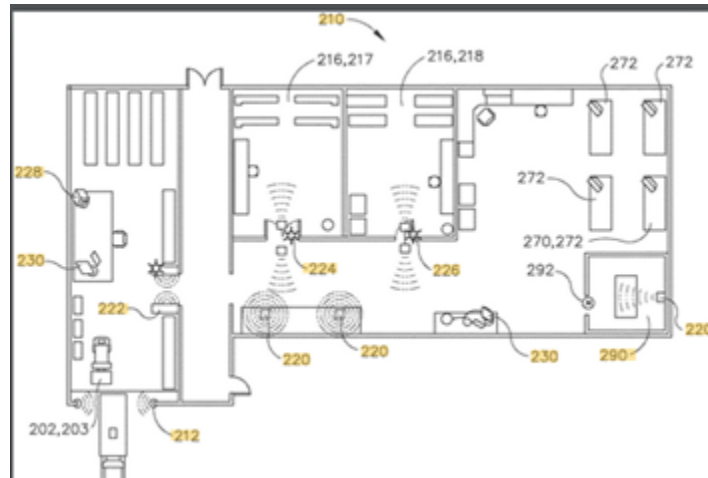
Figure 15

Figure 15 is a depiction of the lab management inventory sharing system designed by Barnes et al. (2023).

The system is used to effectively share consumables amongst various laboratory instruments. The system consists of middleware software run on a processor which controls the various instruments in the system. The consumables are designed in such a way that they are readily removeable and attachable to instruments at ones convenience. Traditional inventory management is significantly improved because resource sharing can be enacted at various points to reduce waste. The middleware software allows for easy communication between the instruments and the inventory database.

Figure 15

System and Method for Inventory Sharing in a Laboratory Management System (Barnes et al., 2023)



Evaluation System

The evaluation process was conducted through a survey using qualitative method in determining the performance of the project. The survey was based on the TUP evaluation system provided with a set of criteria as follows:

Functionality pertains to ease of operation, provision of comfort and convenience.

Aesthetics of the project refers to color appeal, attractiveness and appropriateness of the design.

Workability pertains to the availability of materials, availability of technical expertise, and availability of tools.

Durability of the project refers to the quality of materials, quality of workmanship, and quality of design.

Economy is evaluated in terms of material needed, time labor spent, and machine required.

Safety pertains to absence of toxics and hazardous materials, absence of sharp edges and provisions of protective devices.

Salability/Marketability concerns the consumer's demand, accessibility of the finished product and price affordability.

The initial stage includes the gathering of information about the existing machines which can be applied in the design of Automated laboratory tools inventory systems. The materials that are needed such as: Raspberry Pi 4 Model b and ESP32

The 2nd stage highlights the preliminary design of the prototype by the use of Computer Aided Drafting, the programming of the raspberry pi and the installation of the sensor and wiring which will be connected to the tools.

The prototype will be tested to the actual users, getting their feedback and using their feedback to perform data analysis. The analysis will determine if the prototype is reliable and ready to use.

Chapter 3

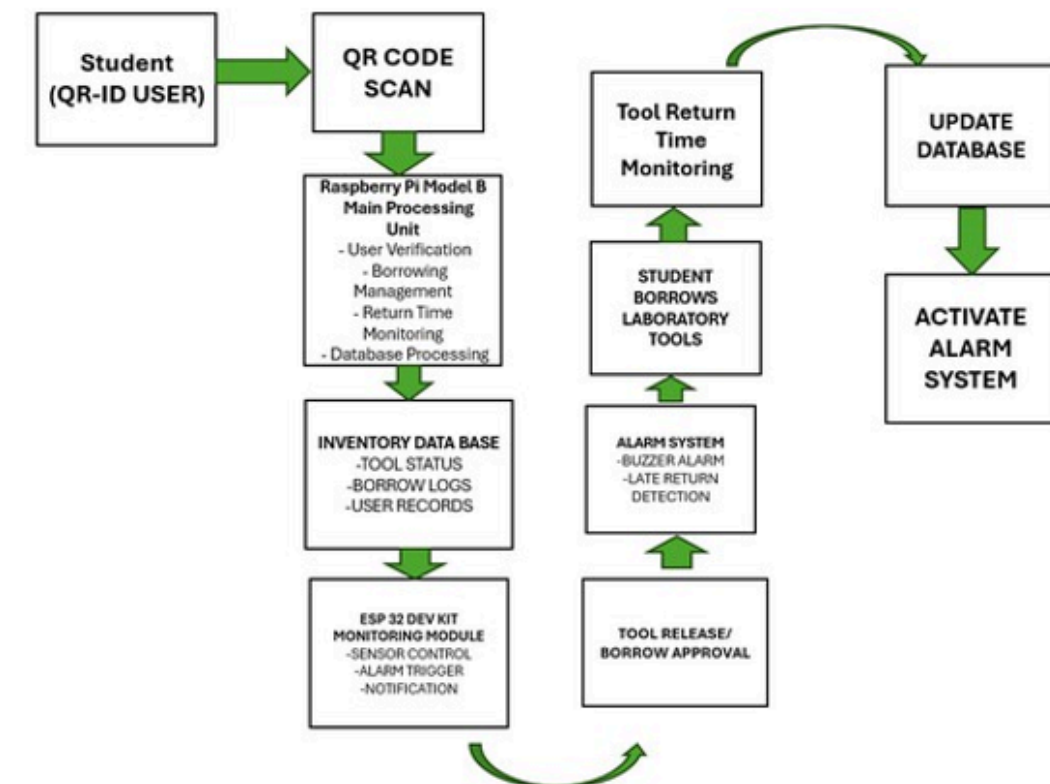
RESEARCH METHODOLOGY

This section includes the designs, frameworks, materials, methods, procedures, and processes used in realizing the objectives of the capstone study. It also describes what resources, activities, strategies, or techniques that were conducted to translate the design into a tangible output.

The block diagram shows the overall process and functionality of the Development of an Automated Laboratory Tools Inventory System. The system is designed to automate the borrowing and returning of laboratory tools using QR code scanning and getting the data of students who borrow the tools.

Figure 16

Block Diagram



Raspberry Pi Model B and an ESP32 Dev Kit is responsible for recording the information and also for controlling the systems.

The process begins when a student scans their ID QR code for identification using the QR code scanner. The scanned information will be sent to the Raspberry Pi Model B, which serves as the main processing unit of the system. The Raspberry Pi verifies the student's id number, manages borrowing transactions, monitors the return schedule of laboratory tools, and processes the inventory database records. The inventory database stores important information such as tool status, borrow logs, student records, due dates and return records.

Once the student is verified, the system grants tool release or borrowing approval. The ESP32 Dev Kit functions as the monitoring and communication module of the system. This performs functions like sensor control, notification processing, and alarm generation. The communication between the ESP32 and Raspberry Pi is made possible to facilitate real time monitoring of the tools lent out by students. Upon returning out of the laboratory tool, the system monitors its return time. The tool should be returned on the given time amount of 3 hours (3 Hours are the designated time for laboratory activities). The system will update the database with regards to successful return of the tool, however, if the tool is not returned on time designation, an alarm will be triggered and the admin will be notified that the tools are not returned yet.

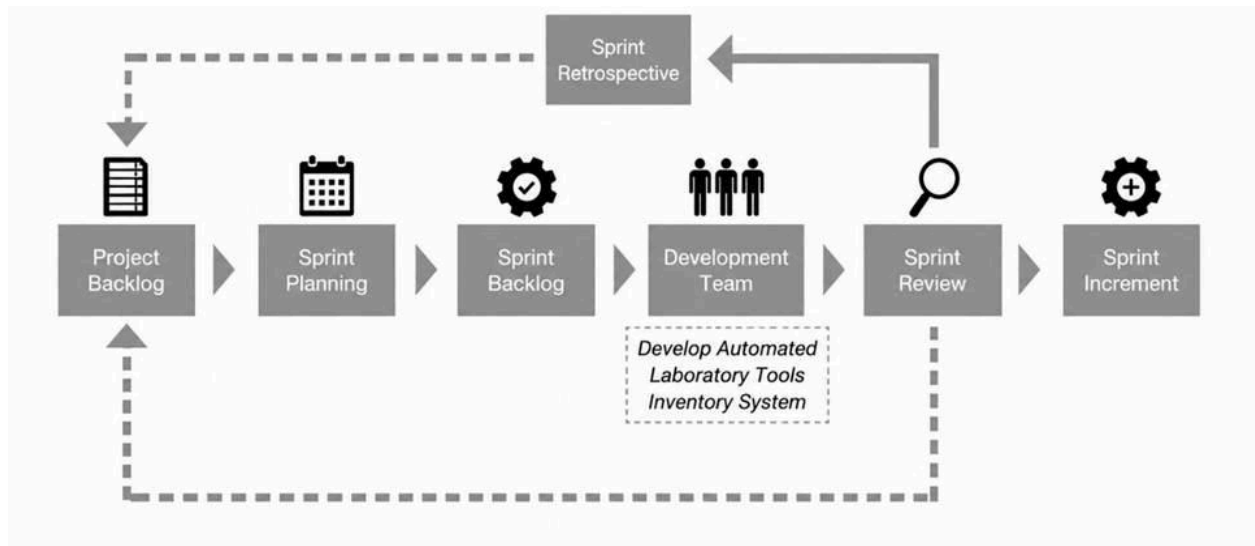
Figure 19

Website App Operational Flowchart

1. *SCANNING STUDENT NUMBER*
2. *RASPBERRY STORING INFORMATION VERIFIED*
 - *SCANNING THE QR OF TOOLS – TO OPEN THE DISIGNATED COMPARTMENT*
 - *RASPBERRY RECORD THE LIMITED TIME FOR GIVEN LABORATORY HOURS*
 - DENIED*
 - *NO ACCESS FOR BORROWING TOOLS*
 -
1. *ESP32 WILL NOTIFY ONCE THE TOOL CANNOT RETURN IN THE COMPARTMENT*

Figure 20

Agile Scrum Methodology



Initially, the researchers contributed to the project backlog by specifying the objectives, system requirements, and hardware and software components necessary for the development of the Automated Laboratory Tools Inventory System. These included the identification of existing inventory management issues, such as manual data recording, data loss, and inefficient tracking of laboratory tools.

Then, they conducted sprint planning to define the project scope, related studies, methodologies, system features, and testing and evaluation procedures. During this phase, the researchers also agreed on the specific technologies to be utilized, including the Raspberry Pi 4 Model B and ESP32 devkit as the primary hardware components, as well as the software frameworks required for database management and system integration.

Next, the researchers divided the tasks among all members of the development team according to their individual strengths and technical capabilities. They established timelines, project milestones, resource allocation, possible risks, and contingency measures to ensure the efficient completion of the system.

Afterward, the sprint backlog was prepared, which outlined the specific tasks, deliverables, responsibilities, and target deadlines for each development phase. Afterwards, the researchers carefully executed the project objectives through continuous coordination with technical advisers, laboratory personnel, and potential system users. Initially, they designed the hardware architecture of the system, followed by the software development process for the automated monitoring and data recording functions. Iterative development and testing were conducted to ensure system reliability, accuracy, and functionality. For every development cycle, the researchers reviewed each output through sprint reviews, allowing necessary improvements and adjustments to be implemented. This process ensured that the Automated Laboratory Tools Inventory System effectively addressed issues related to data loss, manual tracking, and inefficient laboratory tool management while providing a secure and automated inventory solution.

Figure 21

Testing Contraption Flowchart

The researchers conducted a series of test to evaluate the functionality and performance of the prototype. The overall performance of the said project will be evaluated and reorganize if the system has more than 50% error than the desired performance level. The researchers performed multiple randomized test scenarios to assess the system's capability in monitoring tool withdrawals, recording user information, tracking tool return time, and ensuring data accuracy throughout the inventory process.

The testing procedure began by setting the initial system configuration and test variables, monitoring of laboratory tools, facial recognition of users, and calibration of the Raspberry Pi 4 Model B and ESP32 microcontroller components. Afterwards, test are conducted for the prototype to simulate the actual laboratory testing such as, borrowing of tools, return of tools after being used, recording of data through the sensors, test and limitations of Raspberry Pi and ESP32, and other specifics that the system will be requiring tests. The result will show if the is efficient and reliable.

The Summary results were categorized into 4system responses: (0) All Tests Passed, which indicates successful execution of all inventory functions; (1) Minor Issues Detected, referring to small delays or non-critical synchronization inconsistencies; (2) Major Issues Detected, indicating significant errors in tool tracking or database recording; and (3) System

Failure/Critical Errors, which represent severe hardware or software malfunctions requiring immediate corrective action.

However, if the system has frequent errors more than its supposed performance, the prototype will undergo through reprogramming of data collection and other maintenance if necessary. Researchers conducted analytical process in collecting the data by the use of Analysis of Variance (ANOVA) with the help of Statistical Package for the Social Sciences (SPSS) with the version 25 so that the statistical consistency will be determined and the performance reliability of the system.

The F-statistic computed using the following Equations:

F_s = F-statistic; MST_{bg} = Mean square treatment between groups;

MSE_{wg} = Mean square error within groups;

X_{xy} = observed system response value;

T_x = total score of group x; G_t = grand total of all observations;

n_x = number of observations in group x; and

N = total number of observations.

$$F_s = \frac{MST_{bg}}{MSE_{wg}} \tag{1}$$

$$MST_{bg} = \sum_{x=1}^k \left(\frac{T_x^2}{n_x} \right) - \frac{G_t^2}{N} \tag{2}$$

$$MSE_{wg} = \sum_{x=1}^k \sum_{y=1}^n X_{xy}^2 - \sum_{x=1}^k \left(\frac{T_x^2}{n_x} \right) \tag{3}$$